AI-Enhanced Personalized Healing Environments Using Ayurvedic Principles

Prakriti Ayursensora – Smart Aroma & Healing Environment System

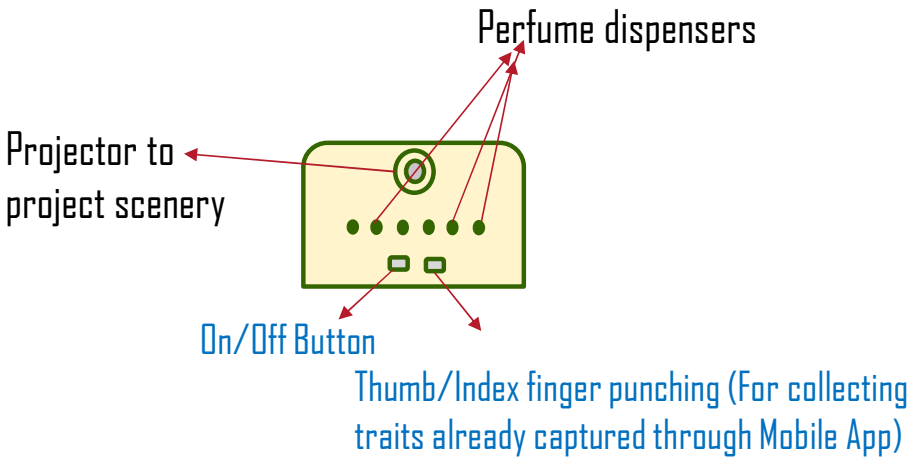
Submitted by:

Dr. Krantee M. Jamdaade | Dr. Harshali Y. Patil | Dr. Ninad D. Mehendale

KJSIM | SSBAS | KJSSE

K J Somaiya Institute of Management, India

Idea : Deluxe and Premium Hospital Wards

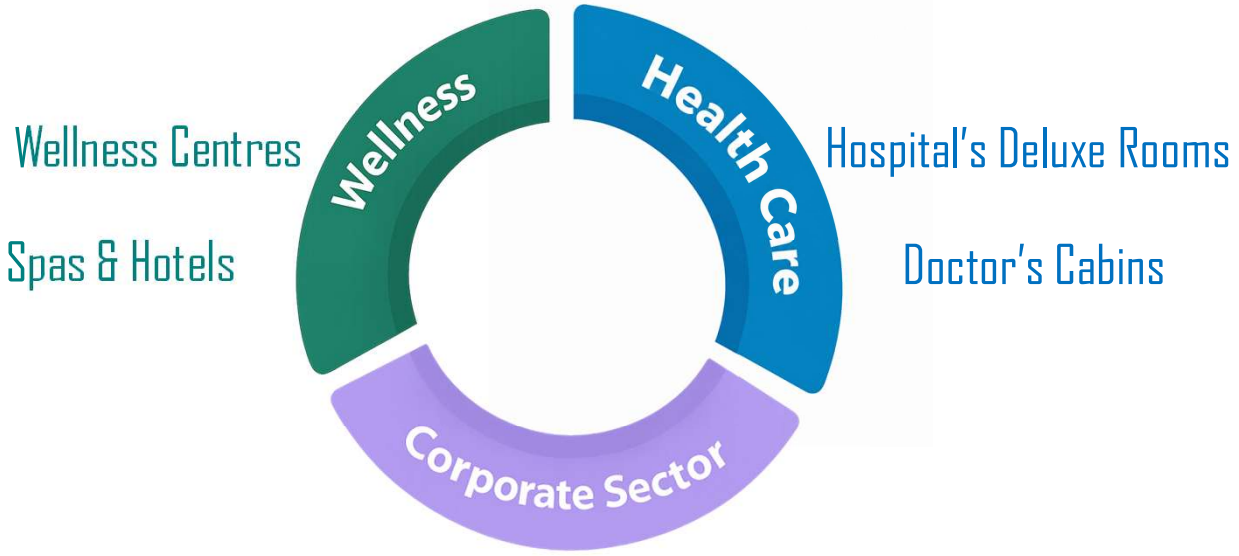




₹ 2,999.00

Note: Camera can be replaced by Mobile App

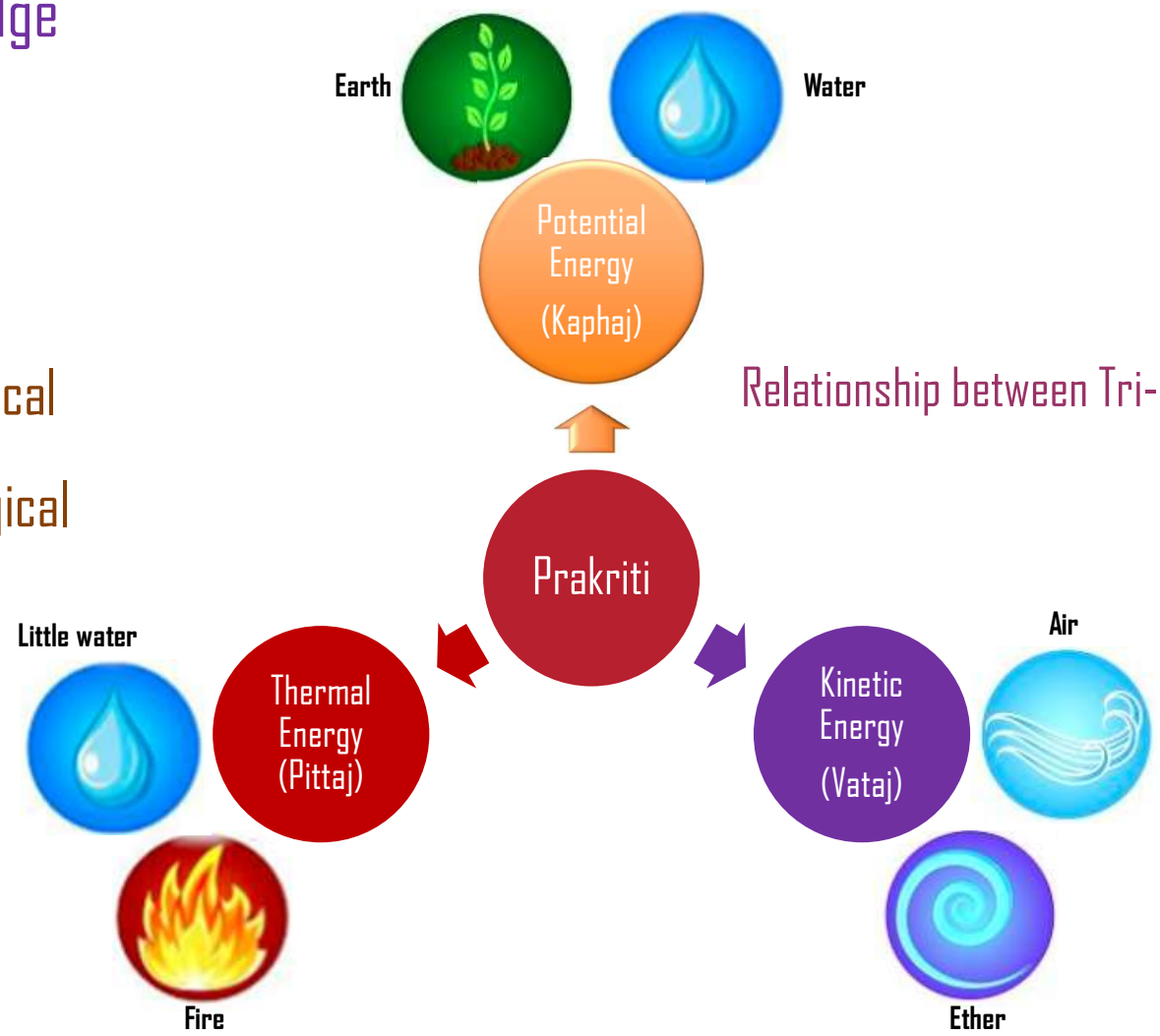
Target User Segments



Indigenous Knowledge

1. Physical
2. Physiological
3. Psychological

Relationship between Tri-Dosha and Five Elements



3 Modules

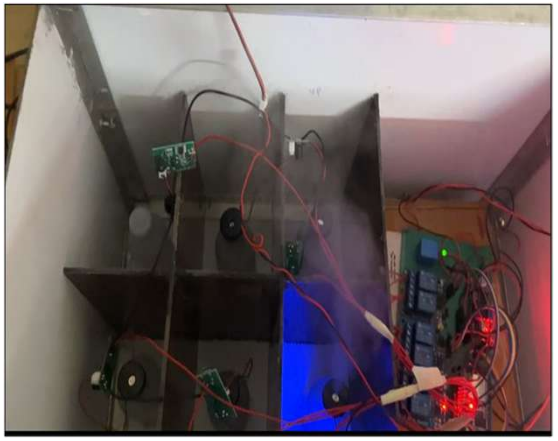
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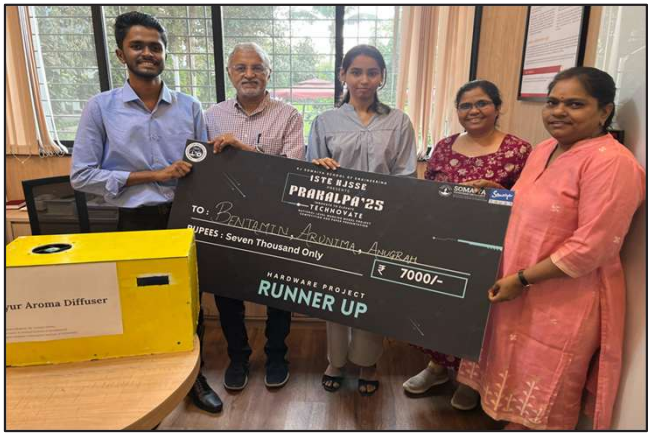
1. Aroma therapy

Kaphaj- Rosemery/ Eucalyptus / Thyme / Peppermint

Pittaj – Rose/ Jasmin / Lavender / Mint

Vataj – Sandal Wood/ Cinnamon/frankincense





3. Music Melody

Kaphaj – Bhairav, Jaunpuri, Hamsadhwani, Desh

Pittaj – Miyan ki Todi, Malkauns, Marwa, Durga

Vataj – Bhimpalasi, Bageshree, Yaman, Kafi

2. Display Technology

Kaphaj- Flowing water, Medows

Pittaj – Mountains + Water bodies

Vataj – Lush Forest

PRAKRITI AYURSENSORA-AI NEBULIZER

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Design Registered



ORIGINAL
क्रम सं/ Serial No. : 236864

INTELLECTUAL PROPERTY INDIA

पेटेंट कार्यालय, भारत सरकार The Patent Office, Government Of India

डिजाइन के पंजीकरण का प्रमाण पत्र | Certificate of Registration of Design

डिजाइन सं. / Design No. : 453172-001

तारीख / Date : 27/03/2025

पारस्परिकता तारीख / Reciprocity Date* :

देश / Country :

प्रमाणित किया जाता है कि संलग्न प्रति में वर्णित डिजाइन जो *NEBULIZER* से संबंधित है, का पंजीकरण, श्रेणी 24-01 में 1.Dr. Krantee Manoj Jamdaade 2. Dr. Harshali Yogesh Patil 3.Dr. Seema Uday Purohit 4.Dr. Vinayak Vitthal Tayade 5.Mr. Benjamin Sebastian 6.Mr. Anugrah Manoj Jamdaade के नाम में उपर्युक्त संख्या और तारीख में कर लिया गया है।

Certified that the design of which a copy is annexed hereto has been registered as of the number and date given above in class 24-01 in respect of the application of such design to *NEBULIZER* in the name of 1.Dr. Krantee Manoj Jamdaade 2. Dr. Harshali Yogesh Patil 3.Dr. Seema Uday Purohit 4.Dr. Vinayak Vitthal Tayade 5.Mr. Benjamin Sebastian 6.Mr. Anugrah Manoj Jamdaade.

डिजाइन अधिनियम, 2000 तथा डिजाइन नियम, 2001 के अध्यधीन प्रावधानों के अनुसरण में।
In pursuance of and subject to the provisions of the Designs Act, 2000 and the Designs Rules, 2001.

जारी करने की तिथि : 10/06/2026
Date of Issue :

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Design Certificate

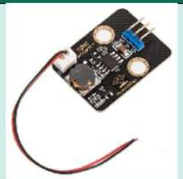
Ayursensora Master Component List

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Prakriti Ayursensora — Master Component List										
AI-Enhanced Personalized Healing Environments Using Ayurvedic Principles | Somaiya Vidyavihar University, Mumbai										
S. No	Component / Item	Image	Category	Purpose / Application	Specs / Notes	Qty	Unit	Unit Cost (₹)	Total Cost (₹)	Link
▸	**AI & CONTROL**									
1	Raspberry Pi 5 (8GB RAM)		Microcontroller / SBC	Main AI hub — runs facial expression analysis, Prakriti inference, soundscape selection and projector output	Quad-core ARM Cortex-A76 @ 2.4GHz, 4GB LPDDR4X, dual HDMI, Wi-Fi, BT 5.0	1	Nos	19,618	19,618	Buy Raspberry Pi 5 Model 8GB RAM online
2	Official Arduino UNO EK R4 WiFi ABX00087_IN – Made in India		Microcontroller	Controls relay switching, nebulizer trigger, fan PWM, and sensor polling	ATmega328P, 14 digital I/O, 6 PWM, USB-B	1	Nos	1,598	1,598	https://robu.in/product/official-arduino-uno-e wifi/
3	L298N Motor Driver Module		Motor Driver	Drives 5V DC fan for mist dispersal with speed control via PWM	Dual H-bridge, 2A per channel, 5–35V motor supply, heatsink included	1	Nos	146	146	Buy L298N Based Motor Driver Module – 2A Online at Robu.in
4	ESP32 Wi-Fi + BLE Module + Screen		Wireless Connectivity	Wireless data bridge between Ayursensora unit and mobile app / dashboard	Dual-core 240 MHz, 802.11 b/g/n, BLE 4.2, 4MB flash	2	Nos	3,832	7,664	ESP32 5 Inch 800×480 IPS TFT Touch WiFi BLE Display

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5	38Pin CP2102 ESP-32 Wifi+Bluetooth Development Board with Type-C USB Interface					1	Nos	355	355	https://robu.in/product/38pin-cp2102-esp-32-wifibluetooth-development-board-with-type-c-usb-interface/
6	SanDisk Extreme PRO microSDXC UHS-I Card		Storage	Stores AI models, Prakriti profiles, aroma schedules and session logs	32GB, UHS-I, Class 10, microSD	2	Nos	749	1,498	https://www.tatacliq.com/sandisk-extreme-microsd-uhs-i-card-512gb-for-4k-video-on-smartphones-and-action-cameras-redyellow/p-mp000000015375497?ef_id=Cj0KCQjwy_fOBhC6ARIsAHKFB79HnPBufQZR7VM4wWABKF22djhSfgGtQ42qhI9WZgGBsyRfFVYdKIaA
▸	**AROMA DIFFUSION SYSTEM**									
7	**Ultrasonic Nebulizer / Mist Maker Module (ACEBOTT)**		Aroma Hardware	Core Ayursensora unit — generates 4–6 micron mist from Ayurvedic essential oils	PWM-controlled piezo disc, 2.4 MHz, –20°C to +70°C, 3.16×2.38 cm	3	722	850	2,550	Buy Now ACEBOTT Ultrasonic Nebuli Humidifier Module Easy-Plug Compatible w ESP32/Arduinozer-humidifier-modu
8	**5V DC Brushless Fan (80mm) — mist dispersal**		Aroma Hardware	Blows atomised aroma mist into the ward environment at controlled speed	5V, 0.25A, 80×80×25 mm, 3-pin connector, PWM speed control	3	Nos	98	294	https://robu.in/product/8025-5v-cooling-fan-with-bolt-nut/
9	5V 4-Channel Relay Module (opto-isolated)		Switching / Control	Independently switches nebulizer, fan, LED strip and audio power under Arduino control	10A / 250V AC contacts, opto-isolated, JQC3F, active-low trigger	2	Nos	143	286	https://robu.in/product/4-channel-isolated-10a-relay-module-opto-coupler-for-arduino-p avr-dsp-a

Ayursensora Master Component List

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10	Helichrysum Essential Oil for Vata (Ectomorph) & Pitta (Mesomorph) Balance | 100% Pure & Therapeutic | 15ml		Aroma Consumable	Therapeutic aroma for Vataj-Pittaj & Pittaj-Vataj Prakriti patients: calming, grounding formulation	10–30 ml, therapeutic grade, GMP-certified supplier	**10**	Bottles	450	4,500	https://www.amazon.in/Flourish-Helichrysum-Essential-Therapeutic-Ayurvedic/dp/B0FPXL8ZN7?th=1
11	Davana Essential Oil (Artemisia Pallens)		Aroma Consumable	Therapeutic aroma for Vataj-Kaphaj & Kaphaj-Vataj Prakriti patients — calming, grounding formulation	10–30 ml, therapeutic grade, GMP-certified supplier	10	Bottles	610	6,100	https://rkaroma.com/products/davana-essenti oil-artemis pallens?variant=43083343921390&country=IN currency=INR&utm_medium=product_sync& m_source=google&utm_content=sag_organic tm_campaign=sag_orga
12	Tea Tree Essential Oil for Pitta (Mesomorph) & Kapha (Endomorph) Balance | 100% Pure & Therapeutic | 15ml		Aroma Consumable	Therapeutic aroma for Pittaj-Kaphaj & Kaphaj-Pittaj Prakriti patients — calming, grounding formulation	10–30 ml, therapeutic grade, GMP-certified supplier	**10**	Bottles	300	3,000	https://www.amazon.in/Forever-Flourish-Tea-Tree-Essential/dp/B0FPXLJSMK?th=1
13	Tridoshic (Sama Prakriti) Balancing		Aroma Consumable	Therapeutic aroma for Pitta dosha patients — cooling, soothing formulation	10–30 ml, therapeutic grade, GMP-certified supplier	**10**	Bottles	499	4,990	https://www.amazon.in/Lotus-Essenti Therapeutic-Ayurved Spiritual/dp/B0FPXMTC4H?th
14	Glass Essential Oil Bottles — 30ml with dropper (spare)		Aroma Consumable	Refillable storage for prepared Prakriti oil blends during pilot study	30ml amber glass, ROPP cap with dropper	**9**	Bottles	9	81	https://dir.indiamart.com/impcat/essential-oil-bottle/capacity-30-ml-q15505384/

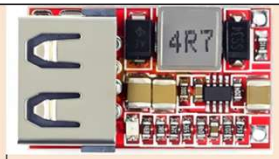
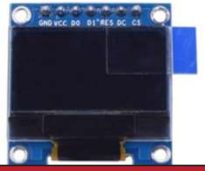
Ayursensora Master Component List

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▸ **BIOMETRIC & USER INPUT**										
15	AS608 Optical Fingerprint Scanner Module		Biometric Sensor	Identifies returning patients by fingerprint to auto-load saved Prakriti profile	Optical 500 dpi, UART TTL, up to 1000 templates, 3.3–5V	1	Nos	879	879	https://robu.in/product/as608-optical-fingerpri sensor-fingerprint-module/
16	Arducam IMX219 Wide Angle Camera Module for NVIDIA Jetson Nano, Raspberry Pi Compute Module CM3, CM3+ and CM4		Biometric Sensor	Real-time facial expression analysis using OpenCV / TensorFlow Lite for emotion-aware Prakriti tuning	OV5647, 1080p@30fps, CSI ribbon, day/night IR-cut, fixed focus	1	Nos	3,568	3,568	https://robu.in/product/arducam-imx219-wi angle-camera-module-for-nvidia-jetson-na raspberry
17	7-inch Capacitive Touchscreen (HDMI + USB)		User Interface	Primary patient-facing interface — Prakriti questionnaire, manual dosha selection, feedback input	800×480, capacitive multi-touch, HDMI video + USB HID, 5V powered	1	Nos	6,333	6,333	https://robu.in/product/7-official-raspberry-pi display-with-capacitive-touchscreen/?gad_source=1&gad_campaignid 9974686076&gbraid=0AAAADvLFWeWOr TVkMkuFYc5UZdEvGq&gclid=CjwKCAjw- dfOBhAjEiwAq0RwI8ICXn8W2IJIoSJwSTd
▸ **POWER SYSTEM**										
18	5V / 10A SMPS Power Supply Unit		Power	Mains power for Raspberry Pi, Arduino, sensors, LEDs and peripherals	Input 90–260V AC, Output 5V / 10A (50W), regulated, short-circuit protected	1	Nos	585	585	https://robodo.in/products/5v-10a-smps-powe supply?variant=42802891849928&country=IN currency=INR&utm_medium=product_sync& m_source=google&utm_content=sag_organic tm_campaign=sag_organic&gad_source=1&g _campaignid=21315610705&gbraid=0AAAA
19	Stuffcool Giga Max Smallest 25000mAh 100W Power Bank, Built-in Type-C Cable, Charges MacBook Pros, laptops, supports Samsung Super		Power	Seamless power continuity during mains failure — keeps unit running during ward power cuts	Input 12V DC, pass-through UPS, auto-switchover, low battery indicator	1	Nos	4,899	4,899	https://www.amazon.in/Stuffcool-Giga-M Smallest-25000mAh/dp/B0FL2H8HZ6?t

Ayursensora Master Component List

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21	5V 3A Buck Converter / DC-DC Step-Down Module		Power	Efficient 12V → 5V step-down for powering RPi5 and peripherals from battery bank	MP2307 / LM2596, input 4–35V, output 1.25–30V adjustable, up to 3A	2	Nos	48	97	https://www.utlinglots.in/products/6v-24v-to-5v-3a-output-dc-dc-step-down-buck-converter-with-u a?gad_source=1&gad_campaignid=21904660 &gbraid=0AAAADijk2WZXbHxEmWsxuU QGjDfSzK&gclid=CjwKCA
▸	**AUDIO-VISUAL OUTPUT**									
22	Portronics Beem 470 Smart LED Projector with 1080p Full HD Resolution, Rotatable Design, Built-in Streaming Apps (Netflix, Prime Video, Hotstar), 4500 Lumens, Screen Mirroring, 5		Visual Output	Projects dosha-specific immersive healing visuals (nature scenes, mandalas, colour therapy) onto ward wall/ceiling via RPi5 HDMI	720p native, 1080p support, ≥1500 lm, HDMI + USB, built-in speaker, portable	1	Nos	7,997	7,997	https://www.amazon.in/Portronics-Project Resolution-Rotatab Streaming/dp/B0D81CWXK4/ref=sr_1_8?cric IWOAB19KPGNK&dib=eyJ2IjoiMSJ9.GI q eUfNza8pekZ5csbezamc4QUArN32RUtUxW
23	Bluetooth / Aux Speaker (10W)		Audio Output	Plays dosha-specific ambient music, binaural beats and healing soundscapes	10W RMS, Bluetooth 5.0 + 3.5mm aux, 80Hz–18kHz	1	Nos	499	499	https://www.amazon.in/dp/B0DG2ZHKCZ/?_e oding=UTF8&aref=df5arToB2G&pd_rd_i=B0 G2ZHKCZ&ref_=sbx_be_s_sparkle_ssd_vid& d=1775665510&pd_rd_w=B2DqA&content- id=amzn1.sym.9269eab1-ae85-443b-9ec2- b2fa4ebaad05%3Aamzn1.sym.9269eab1-ae85-
24	WS2812B Addressable RGB LED Strip (5V, 5m, 60LED/m)		Ambient Lighting	Colour therapy — warm/cool/neutral lighting tuned to Vataj/Pittaj/Kaphaj dosha	5V WS2812B, IP30, 60 LEDs/m, 5m reel, individually addressable	2	Reels	2,594	5,187	https://zbotic.in/product/epoxy-ws2812b-sh led-strip-60-led-meter-flexible-5m-r waterproof- ip65/?gad_source=1&gad_campaignid=20903 6778&gbraid=0AAAAACTNRdV1bOh7BHH ozxlCaKxOd88&gclid=CjwKCA
25	OLED Display Module 0.96" (SSD1306, I2C)		Status Display	Shows active dosha mode, aroma status and sensor readings on the Ayursensora unit	SSD1306, 128×64 px, I2C (0x3C), 3.3–5V	3	Nos	219	657	https://robu.in/product/0-96-oled-display-module/

Methodology

Phase I
System Development & Validation
Conducted at Somaiya Ayurveda Centre in collaboration with technology labs.

Phase II
Clinical Deployment
Full-scale testing at K. J. Somaiya Hospital, comparing personalized healing rooms with standard care.
Minimum 100 participants.

Phase III
Outcome Assessment
Continuous biofeedback and post-discharge follow-up.
Physiological, emotional, and satisfaction-based outcomes using advanced statistical analysis



Proposed Design

Music Melody

Kaphaj – Bhairav, Jaunpuri, Hamsadhwani, Desh

Pittaj – Miyan ki Todi, Malkauns, Marwa, Durga

Vataj – Bhimpalasi, Bageshree, Yaman, Kafi

Aroma therapy

Kaphaj- Rosemery/ Eucalyptus / Thyme / Peppermint

Pittaj – Rose/ Jasmin / Lavender / Mint

Vataj – Sandal Wood/ Cinnamon/frankincense

Display Technology

Kaphaj- Flowing water, Medows

Pittaj – Mountains + Water bodies

Vataj – Lush Forest

360^0 Well Being

{Reduce = Stress , Anxiety, Depression}

Project Timeline

Year 1	Year 2	Year 3	Year 4	Year 5
20%	**45%**	**15%**	**15%**	**5%**
Foundational Research	Prototype & Small Trials	Large Clinical Trials	Commercial Feasibility	Policy Advocacy & Dissemination

Budget

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Budget Head	Estimated Amount (INR)	Justification
Personnel	₹90,00,000	Salaries for PI, Co-PIs, post-doctoral fellows, research associates, technical staff, and admin assistant
Equipment & Infrastructure	₹80,00,000	High-precision sensors, aroma diffuser and nebulizer systems, LED projectors, computing units, software licenses
Clinical Trials & Validation	₹45,00,000	Participant recruitment, compensation, clinical setup, consumables, ethics approvals
Consumables & Materials	₹20,00,000	Essential oils, sensor modules, prototyping components, clinical supplies
Travel & Collaboration	₹15,00,000	Site visits, national workshops, stakeholder meetings, tech demos
Dissemination & IP Filing	₹20,00,000	Publications, training, IPR filing, product launch strategy
Contingency & Overheads	₹30,00,000	Institutional overheads, unforeseen expenses
Total	₹3,00,00,000	

Institutional Support

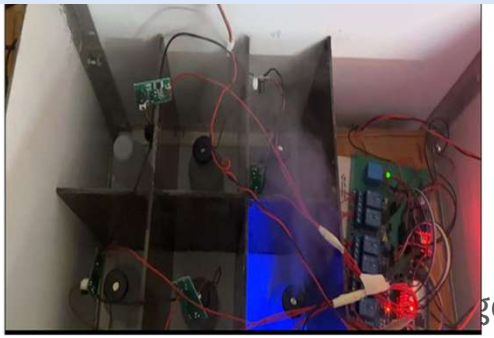


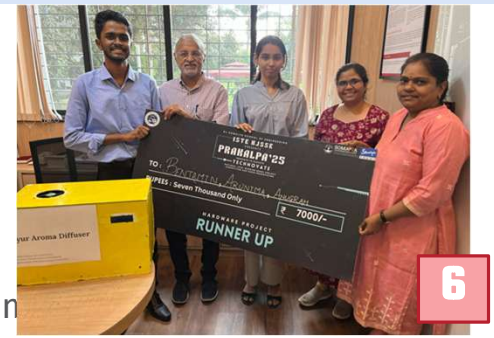
Somaiya Ayur Vihar



KG Mittal Ayurvedic Hospital

Current Status





Our Team



Krantee

Role: Overall project preparation, administration and coordination. Ayurveda principles, Oversees timeline management, milestone tracking, and ICSSR compliance.Coordinates between the Somaiya Ayurveda Centre and K. J. Somaiya Hospital for participant recruitment and implementation.



Ninad

Role: Technical & AI System Lead responsible for the design and implementation of AI-based Prakriti assessment engine. Sensor integration and real-time feedback system. Supervises the development of the hardware prototype, control algorithms, and adaptive system logic.



Harshali

Role: Technical Development & Integration System Lead Responsible for interdisciplinary integration across Ayurveda, AI, psychology, and clinical operations. Oversees Machine Learning, Data Analysis & Dashboard Design



Seema

Role: Clinical Trials & Psychological Assessment Lead who Designs the clinical trial protocols, including: Participant inclusion/exclusion criteria. Ethical approvals and patient consent workflows. Oversees psychological data collection (PSS, HADS, PSQI).



Om

Role: International Public Health and Evaluation Expert who will lead on:Outcome measurement protocols and evidence validation. Ensuring international standards for clinical trials and publication ethics.Brings expertise in global healthcare systems, helping align project outcomes for future WHO or NHS integration.



Vinayak

Role: Ayurveda and Prakriti Expert who Provides Ayurvedic expertise in: Validating Prakriti models. Mapping essential oils, ragas, and visual elements to dosha types (Vata, Pitta, Kapha). Assists in developing standardized Prakriti questionnaires and Ayurvedic training for other team members.

AACSB ACCREDITED

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Thank You

simsr.somaiya.edu

राग चिकित्सा – कौन सी बीमारी में कौन सा राग? 🎵

(भारत में सबसे आम बीमारियाँ – Part 1)

TABLE 1 :

बीमारी	राग	कब सुनें	असर
तनाव / Stress और Anxiety	राग भैरवी राग दरबारी	सुबह या रात को सोने से पहले (30–45 min)	Cortisol (stress hormone) कम होता है मन शांत होता है
उच्च रक्तचाप (High BP)	राग भूपाली राग मालकौंस	शाम को सूरज ढलने के बाद (30 min)	Heart rate और BP धीरे-धीरे normal होता है
नींद न आना (Insomnia)	राग दरबारी राग बागेश्री	रात को सोने से 30 मिनट पहले (headphones से)	Brain को deep relaxation signal मिलता है
अवसाद (Depression)	राग भैरवी राग यमन	सुबह और शाम दोनों (20–30 min)	Serotonin और Dopamine release होता है
सिरदर्द / Migraine	राग तोड़ी राग भैरवी	दर्द होने पर शांत जगह में (20 min)	Nerve tension कम होती है दर्द में राहत मिलती है

★ ध्यान दें :

→ यह जानकारी Raga Chikitsa (पारंपरिक भारतीय संगीत चिकित्सा) और आधुनिक research पर आधारित है

→ राग को सुनते वक्त – आँखें बंद, शांत जगह, headphones best है

Like करो –
अगर यह नई जानकारी थी ❤

Sapta Swar Academy